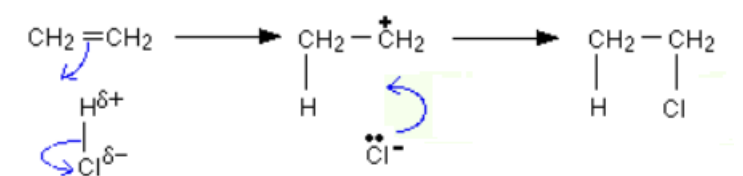
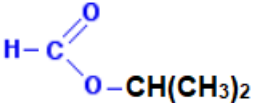


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13	(a)	(i)		radical substitution (1) (electrophilic) addition (1)	2			2		
		(ii)		award (1) for either of following because only one product is formed because the reaction between ethane and chlorine / radical substitution has lots of possible products			1	1		
		(iii)		 dipole and curly arrow on H—Cl molecule (1) curly arrow from the double bond and from the lone pair/negative charge (1) carbocation structure (1) product formed on right hand side (1)		4		4		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$n(\text{CH}_3\text{COOH}) = \frac{25}{60} = 0.417 \text{ mol (1)}$ $n(\text{CH}_3\text{COOCH}_3) = \frac{34}{100} \times 0.417 = 0.142 \text{ mol (1)}$ mass of $\text{CH}_3\text{COOCH}_3 = 0.142 \times 74 = 10.5 \text{ g (1)}$		3		3	2	
		(ii)		bulb of thermometer near exit to condenser (1) water flow from bottom to top of condenser (1)	2			2		2
		(iii)		award (1) for either of following sulfuric acid (is a dehydrating agent and) removes water (from the right-hand side of the equation) sulfuric acid increases H^+ concentration (on the left-hand side of the equation) the equilibrium position shifts to the right-hand side (to oppose the change) (1)			2	2		
	(c)					1		1		
				Question 13 total	4	8	3	15	2	2